

NexRay AI

Medical Assistant Platform • Clinical Decision Support Report

25 July 2026 | 01:41 PM

Session ID: 24	Modules Used: Symptom Checker • Refined Diagnosis	Date: 25 July 2026 01:41 PM
----------------	---	-------------------------------

Symptom Analysis

Possible Condition	Confidence
COVID-19 (SARS-CoV-2)	92%
Influenza (Flu)	78%
Malaria with secondary respiratory infection	45%

■ **Urgency Level: Urgent — Same day attention required**

Recommended Tests

- COVID-19 RT-PCR or rapid antigen test (nasopharyngeal/oropharyngeal swab)
- Malaria rapid diagnostic test (RDT) or blood smear
- Full blood count (FBC) with differential
- Chest X-ray (if shortness of breath is significant)
- Influenza rapid test (if COVID-19 negative and clinical suspicion remains high)

Suggested Treatment

- If COVID-19 confirmed: supportive care including rest, hydration, paracetamol or ibuprofen for fever and body aches; monitor oxygen saturation closely given shortness of breath
- If malaria co-infection suspected: commence appropriate antimalarial therapy (artemisinin-based combination therapy preferred in West Africa)
- Manage respiratory symptoms: monitor oxygen saturation, consider oxygen supplementation if SpO2 <94%, assess for need for escalated care
- Consider empiric broad-spectrum antibiotic coverage if secondary bacterial infection suspected based on clinical deterioration

Next Steps

- Prioritize COVID-19 testing immediately to guide isolation precautions
- Perform malaria screening given West African context and fever presentation
- Monitor respiratory status closely; document oxygen saturation and respiratory rate
- Obtain chest imaging if shortness of breath worsens or does not improve with supportive care
- Review test results and adjust treatment plan accordingly within 24 hours

Summary: Patient presents with a constellation of symptoms highly suggestive of COVID-19, particularly given the specific triad of fever, dry cough, and loss of taste/smell. Influenza remains a differential, and malaria should not be excluded in the West African context. The presence of shortness of breath elevates clinical concern and warrants prompt respiratory assessment. Immediate testing for COVID-19 and malaria is recommended, with supportive care and close monitoring of respiratory status.

Refined Diagnosis — Based on Test Results

Final Diagnosis: Right lower lobe pneumonia with secondary parapneumonic pleural effusion

■ Confirmed Conditions

- Pleural Effusion (Right side)** (Confirmed) — Confirmed by ultrasound showing 400ml fluid, CT imaging and lateral chest X-ray
- Pneumonia (Right lower lobe)** (Confirmed) — CT shows right lower lobe consolidation, elevated WBC 15,200 and CRP 98 support active infection

■ Ruled Out

- Atelectasis** — CT imaging showed consolidative pattern rather than collapse, inflammatory markers support pneumonia

■ **Urgency Level: Urgent — Same day attention required**

Updated Treatment

- Broad-spectrum antibiotics: amoxicillin-clavulanate or ceftriaxone
- Consider azithromycin to cover atypical organisms
- Oxygen therapy to maintain SpO2 above 90%
- Serial chest imaging in 5-7 days to assess response

Next Steps

- Repeat blood cultures if fever persists after 48-72 hours of antibiotics
- Repeat chest X-ray at 5-7 days to monitor resolution
- Consider TB screening if no improvement in 7-10 days

***Summary:** Pneumonia with secondary pleural effusion confirmed by imaging and inflammatory markers. Antibiotics initiated.*

■ **DISCLAIMER:** This report is generated by NexRay AI and is intended solely as a clinical decision-support tool. All findings, conditions, treatments, and recommendations are AI-generated suggestions and do not constitute a formal medical diagnosis or prescription. The clinical judgment of the attending medical professional must be applied before any action is taken.